

The Inverse-GCD Operator Q_N (Q6): Definitions, a Restricted Rayleigh Bound, and an Exploratory Spectral Programme

Crown-jewel operator note — what is proved and what is open

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Scope. Q6 is developed here as a spectral operator on $\{1, \dots, N\}$. Proved: Bridge* on single prime pairs, multi-representation Goldbach vectors, and the nonnegative cone. False as stated: universal full-spectrum floors $\lambda_{\min} > -1/2$. No Navier–Stokes claim.

Status

Claim	Status
Definitions of Q_N , $\tilde{Q}_N$, H_N	Locked
Bridge*: $R(e_p - e_q) > -1/2$ on $\tilde{Q}_N$	Proved
Positive orthant: $v \geq 0 \Rightarrow v^\top \tilde{Q}_N v \geq 0$	Proved
$\lambda_{\min}(Q_N) > -1/2$ for all N	False
$\lambda_{\min}(\tilde{Q}_N) > -1/2$ for all N	False
Dark-state $\Leftrightarrow$ Goldbach (June 2026 draft)	Withdrawn
Multi-representation Bridge*	Proved

Abstract

We give a corrected account of the inverse-GCD operators used in earlier “Q6” drafts. Write

$$Q_N(i, j) = \frac{1}{\gcd(i, j)}, \quad \tilde{Q}_N(i, j) = \frac{1}{\gcd(i, j)\sqrt{ij}}, \quad 1 \leq i, j \leq N,$$

and let H_N be the degree-normalized form of $\tilde{Q}_N$. We prove Bridge* for single prime pairs $v = e_p - e_q$ and for multi-representation vectors $v_k = \sum_{p \in \mathcal{R}(k)} (e_p - e_{k-p})$ (even k): both satisfy $R > -1/2$ on $\tilde{Q}_N$. The multi-rep case reduces to the pair case by a nonnegative cross-term identity. We also record that every nonnegative vector has nonnegative Rayleigh quotient on $\tilde{Q}_N$. We withdraw the full-spectrum Bridge claim $\lambda_{\min} > -1/2$: already for moderate N , $\lambda_{\min}(Q_N) < -1/2$ and $\lambda_{\min}(\tilde{Q}_N) < -1/2$. Earlier dark-state / Goldbach packaging and any Navier–Stokes claim from these matrices are withdrawn.

1 Operators

Definition 1.1 (Raw inverse-GCD).

$$Q_N(i, j) = \frac{1}{\gcd(i, j)}, \quad 1 \leq i, j \leq N.$$

Definition 1.2 (Normalized inverse-GCD).

$$\tilde{Q}_N(i, j) = \frac{1}{\gcd(i, j)\sqrt{ij}}.$$

In particular, for distinct primes $p, q \leq N$,

$$\tilde{Q}_N(p, p) = \frac{1}{p^2}, \quad \tilde{Q}_N(p, q) = \frac{1}{\sqrt{pq}}.$$

Definition 1.3 (Degree-normalized form). Let $d_i = \sum_{j=1}^N \tilde{Q}_N(i, j)$ and $D = \text{diag}(d_1, \dots, d_N)$. Set

$$H_N = D^{-1/2} \tilde{Q}_N D^{-1/2}.$$

Remark 1.4 (Hygiene). These three matrices are different. A spectral statement about one does not automatically transfer to another, nor to any fluid commutator called “Theorem H” in companion notes.

Remark 1.5 (Withdrawn identity). Some June 2026 drafts asserted $1/n = \sum_{d|n} \mu(d)\varphi(d)/d^2$. This fails already at $n = 2$ ($1/2$ versus $3/4$). No argument in this note uses that identity.

2 What fails: full-spectrum floors

Proposition 2.1 (Full-spectrum Bridge is false). There exist N such that

$$\lambda_{\min}(Q_N) < -1/2 \quad \text{and} \quad \lambda_{\min}(\tilde{Q}_N) < -1/2.$$

In particular the universal claims $\lambda_{\min}(Q_N) > -1/2$ and $\lambda_{\min}(\tilde{Q}_N) > -1/2$ for all N are false.

Numeric certificate. Direct eigendecomposition yields, for example, $\lambda_{\min}(Q_{10}) \approx -1.90 < -1/2$ and $\lambda_{\min}(\tilde{Q}_{20}) \approx -0.505 < -1/2$. Reproducible by the accompanying script `scripts/bridge_floor_verify.py`. $\square$

Remark 2.2. A claimed universal lower bound $\lambda_{\min}(H_N) \geq -3/14$ is likewise not uniform in small N ; see the companion verification suite. We do not package any H_N floor as a theorem here.

3 Bridge*: single prime pair

Theorem 3.1 (Bridge*, single pair). Let $p \neq q$ be primes and $N \geq \max(p, q)$. For $v = e_p - e_q$,

$$R(v) := \frac{v^\top \tilde{Q}_N v}{\|v\|_2^2} = \frac{1}{2} \left(\frac{1}{p^2} + \frac{1}{q^2} \right) - \frac{1}{\sqrt{pq}} > -\frac{1}{2}.$$

Proof. Since $\|v\|_2^2 = 2$,

$$v^\top \tilde{Q}_N v = \tilde{Q}_N(p, p) + \tilde{Q}_N(q, q) - 2\tilde{Q}_N(p, q) = \frac{1}{p^2} + \frac{1}{q^2} - \frac{2}{\sqrt{pq}},$$

hence

$$R = \frac{1}{2} \left(\frac{1}{p^2} + \frac{1}{q^2} \right) - \frac{1}{\sqrt{pq}}.$$

Then

$$R + \frac{1}{2} = \frac{1}{2} + \frac{1}{2p^2} + \frac{1}{2q^2} - \frac{1}{\sqrt{pq}} > \frac{1}{2} - \frac{1}{\sqrt{pq}}.$$

Distinct primes force $pq \geq 6$, so $1/\sqrt{pq} \leq 1/\sqrt{6} < 1/2$. Therefore $R + 1/2 > 0$. $\square$

Remark 3.2. *The quotient may be negative: $(p, q) = (2, 3)$ gives $R \approx -0.228 > -1/2$. The bound is a restricted floor on difference vectors of single prime pairs, not a full-spectrum eigenvalue floor.*

4 Positive orthant

Proposition 4.1 (Nonnegative cone). *If $v \in \mathbb{R}^N$ satisfies $v_i \geq 0$ for all i , then $v^\top \tilde{Q}_N v \geq 0$.*

Proof. Every entry of $\tilde{Q}_N$ is positive, so the quadratic form on the nonnegative orthant is a sum of nonnegative terms. $\square$

5 Earlier drafts — corrected status

Earlier public drafts pushed Q6 toward Strong Goldbach via a dark-state lemma and a full-spectrum Bridge floor, and toward RH / Navier–Stokes packaging. Those extensions are not supported by the arguments as written; they are marked exploratory-failed below and are not part of the proved core.

1. **Broken detector.** The indicator difference $v_k(j) = 1_{\mathbb{P}}(j) - 1_{\mathbb{P}}(k - j)$ vanishes on genuine Goldbach pairs $p + (k - p) = k$, so it does not detect them.
2. **False sign lemma.** On raw Q_N , for $v = e_p - e_q$, $\langle v, Q_N v \rangle = 1/p + 1/q - 2$ is typically negative (e.g. $(3, 7) \mapsto -1.52$). A claimed equivalence “dark state $\Leftrightarrow$ no Goldbach” does not hold as written.
3. **No NS map.** A spectral floor on Q_N , $\tilde{Q}_N$, or H_N is not, by itself, a bound on $\|(u \cdot \nabla)u\|$ for Leray–Hopf solutions on $\mathbb{T}^3$.

Older DOIs may remain as dated history. The public credit for this operator is Theorem 3.1, Theorem 6.2, and Propositions 2.1–4.1.

6 Bridge*: multi-representation

Lemma 6.1 (Cross-term factorization). *For distinct primes $p \neq q$ and $r \neq s$,*

$$\tilde{Q}_N(p, r) - \tilde{Q}_N(p, s) - \tilde{Q}_N(q, r) + \tilde{Q}_N(q, s) = \left(p^{-1/2} - q^{-1/2} \right) \left(r^{-1/2} - s^{-1/2} \right).$$

In particular, if $p < q$ and $r < s$, the cross term is strictly positive.

Proof. Each off-diagonal entry is $(\cdot)^{-1/2}$ for distinct primes, so the left-hand side factors as claimed. $\square$

Theorem 6.2 (Bridge*, multi-representation). *Let $k \leq N$ be even and*

$$\mathcal{R}(k) = \{p \text{ prime} : 2 \leq p < k/2, k - p \text{ prime}\}.$$

Set $v_k = \sum_{p \in \mathcal{R}(k)} (e_p - e_{k-p})$ (indices $\leq N$). If $v_k \neq 0$, then

$$R(v_k) = \frac{v_k^\top \tilde{Q}_N v_k}{\|v_k\|_2^2} > -\frac{1}{2}.$$

Proof. Distinct unordered Goldbach pairs for fixed k are pairwise disjoint as sets of prime sites: a shared prime forces the same complementary prime. Write $v_k = \sum_{a=1}^m v_a$ with $v_a = e_{p_a} - e_{q_a}$, $p_a < q_a = k - p_a$. Then $\|v_k\|_2^2 = \sum_a \|v_a\|_2^2 = 2m$ and

$$v_k^\top \tilde{Q}_N v_k = \sum_a v_a^\top \tilde{Q}_N v_a + 2 \sum_{a < b} v_a^\top \tilde{Q}_N v_b.$$

Each diagonal block equals $\|v_a\|_2^2 R(v_a)$ with $R(v_a) > -1/2$ by Theorem 3.1. Each cross block equals the factorization in Lemma 6.1 with $p_a < q_a$ and $p_b < q_b$, hence is positive. Therefore

$$R(v_k) \geq \frac{\sum_a \|v_a\|_2^2 R(v_a)}{\sum_a \|v_a\|_2^2} \geq \min_a R(v_a) > -\frac{1}{2}.$$

$\square$

Remark 6.3. *The worst multi-rep Rayleigh through $N = 200$ coincides with a single-pair case (e.g. $k = 8$, pair $(3, 5)$, $R \approx -0.183$). Richer multi-rep vectors sit strictly higher because of positive cross terms.*

Reproducibility

```
python3 scripts/bridge_floor_verify.py 50
python3 -m unittest tests.test_bridge_star_h_n -v
```

Acknowledgments

This note supersedes the earlier Q6 preprint (10.5281/zenodo.20405589) as the author's intended public write-up of the inverse-GCD operator.

References

- [1] J.R. Simons, *Earlier Q6 draft* (archive), Zenodo 10.5281/zenodo.20405589.
- [2] J.R. Simons, *Bridge* proof draft*, repository note docs/BRIDGE-STAR-PROOF.md (2026).